REPORT

Neural Networks and Fuzzy Systems (EL 6009)

An End-to-End Machine Learning Framework for Heart Disease Prediction Using Clinical Data

Submitted by

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Abstract

Cardiovascular diseases (CVDs) remain the leading cause of global mortality. Early detection and accurate diagnosis are critical for effective intervention and improved patient outcomes. This project, CardioAI, presents a comprehensive, end-to-end machine learning system designed to predict the presence of heart disease based on clinical attributes. Utilizing the widely recognized Cleveland Database from the UCI Machine Learning Repository, we conducted extensive data preprocessing, exploratory data analysis, and feature engineering. A series of Deep Neural Network (DNN) architectures were rigorously evaluated against various hyperparameters, including activation functions, layer depth, dropout rates, and optimizers.

The optimal model—a 3-layer DNN utilizing Exponential Linear Unit (ELU) activation, Batch Normalization, and Dropout—achieved an outstanding ROC-AUC score of 92.66% and an accuracy of 87.5%. To ensure real-world applicability, the trained model was deployed via a RESTful Flask API and integrated with a modern, responsive React frontend utilizing a dark-glassmorphism design paradigm, providing clinicians and users with an intuitive diagnostic tool.

Table of Contents

Serial Number	Topic	Page Number
1	Introduction	5
2	Need For The Study	6
3	About The Dataset	7
4	Methodology and Exploratory Data Analysis	8
5	Model Architecture and Experimentation	12
6	Software Implementation	15
7	Results and Evaluation	16
8	Conclusion	17

List of Figures

Serial Number	Title	Page Number
1	Percentage of missing values across the dataset features prior	8
2	Distribution of the engineered binary target variable showing a relatively balanced dataset.	9
3	Correlation heatmap revealing the linear relationships between continuous variables and the target.	10
4	Distributions of key features categorized by the target variable, illustrating the discriminatory power of features like maximum heart rate (thalach) and age.	11
5	Training and validation curves comparing various activation functions.	12
6	Impact of network depth on convergence and validation accuracy.	13
7	Effect of different dropout rates on the validation loss and accuracy.	13
8	Training loss, validation loss, and validation accuracy for the final optimal DNN.	14
9	Comparison of the top 10 neural network configurations based on multiple classification metrics.	14
10	Confusion Matrix for the final Deep Neural Network, highlighting a low false-negative rate.	16

1. Introduction

The diagnosis of heart disease traditionally relies on a combination of patient history, physical examinations, and complex medical tests such as angiography and electrocardiograms (ECG). While effective, these methods can be time-consuming, expensive, and subject to human error or misinterpretation. Artificial Intelligence (AI) and Machine Learning (ML) offer a complementary approach by identifying non-linear patterns within clinical data to support medical decision-making.

This study aims to leverage deep learning techniques to create a predictive model capable of identifying heart disease using a subset of 14 clinical features. By transitioning from a raw dataset to a fully deployed web application, this project demonstrates not only the statistical efficacy of neural networks in healthcare but also the software engineering practices required to deploy these models into accessible, user-friendly tools.

2. Need For The Study

Despite significant advancements in medical technology, heart disease diagnosis often occurs in advanced stages due to the asymptomatic nature of early-stage cardiovascular issues. The necessity of this project arises from several key healthcare challenges:

1. Early Intervention and Prevention: Automated predictive models can serve as an early-warning system for at-risk patients, identifying subclinical patterns that human practitioners might overlook.
2. Resource Optimization: By accurately predicting risk using non-invasive clinical data (like age, cholesterol, and resting blood pressure), healthcare systems can reduce unnecessary, expensive invasive procedures such as fluoroscopy or angiography.
3. Clinical Decision Support: A high-accuracy ML model acts as a secondary diagnostic tool for clinicians, reducing diagnostic variance, mitigating cognitive bias, and minimizing human error.
4. Accessibility: Deploying the model as a web application democratizes access to advanced diagnostic tools, allowing remote clinics or smaller hospitals to leverage cutting-edge AI without needing specialized local infrastructure.

3. About The Dataset

3.1 Context

The dataset utilized is the multivariate Heart Disease dataset from the UCI Machine Learning Repository, specifically focusing on the Cleveland database. While the original database contains 76 attributes, all published machine learning studies rely on a subset of 14 core attributes. The primary predictive task is binary classification: determining whether a patient has heart disease (target > 0) or does not (target = 0).

3.2 Dataset Attributes

The dataset is composed of 14 clinical, demographic, and physiological attributes:

- id: Unique identifier for each patient.
- age: Age of the patient in years.
- origin / dataset: Place of study.
- sex: Biological sex (Male/Female).
- cp (Chest Pain Type): Categorized as typical angina, atypical angina, non-anginal, or asymptomatic.
- trestbps (Resting Blood Pressure): Measured in mm Hg upon hospital admission.
- chol (Serum Cholesterol): Measured in mg/dl.
- fbs (Fasting Blood Sugar): Boolean indicator if fasting blood sugar > 120 mg/dl.
- restecg (Resting ECG): Results categorized as normal, stt abnormality, or lv hypertrophy.
- thalach: Maximum heart rate achieved during exercise.
- exang (Exercise-Induced Angina): Boolean indicator (True/False).
- oldpeak: ST depression induced by exercise relative to rest.
- slope: The slope of the peak exercise ST segment.
- ca: Number of major vessels (0-3) colored by fluoroscopy.
- thal: Categorized as normal, fixed defect, or reversible defect.
- num (Target): The predicted attribute (angiographic disease status).

3.3 Acknowledgement

Creators: Hungarian Institute of Cardiology, University Hospital (Zurich), University Hospital (Basel), V.A. Medical Center (Long Beach and Cleveland Clinic Foundation).

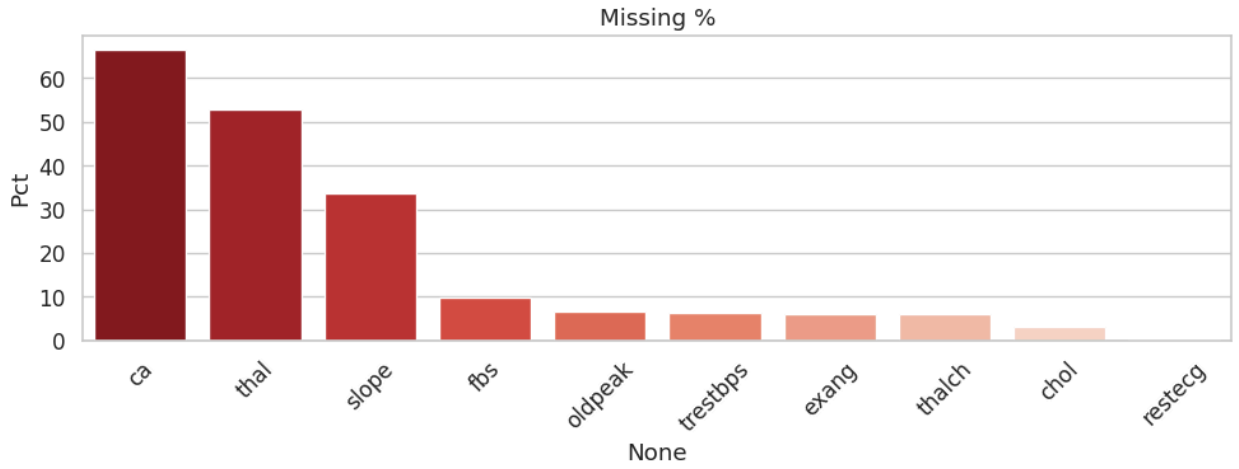
4. Methodology and Exploratory Data Analysis

The project was executed in a structured, multi-phase pipeline encompassing data science, backend engineering, and frontend development.

4.1 Data Processing And Cleaning

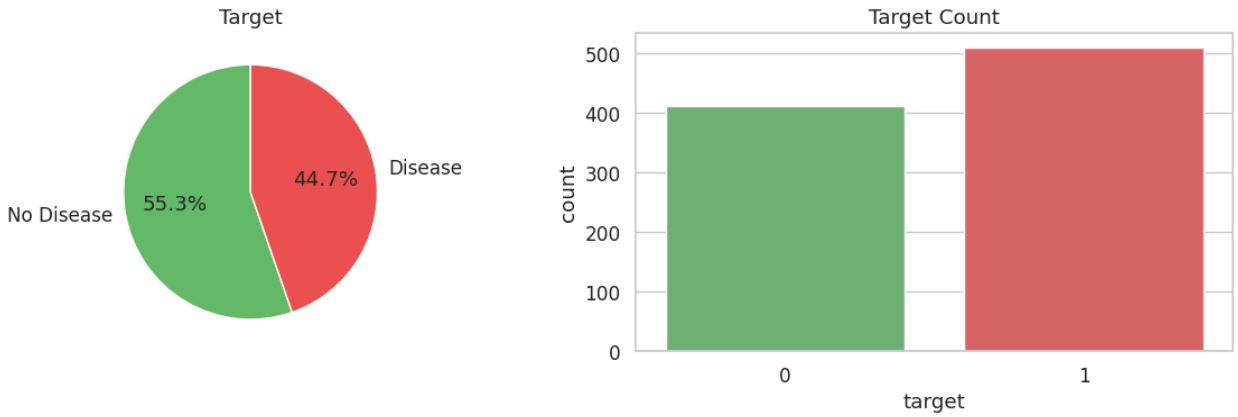
Real-world medical data is often noisy and incomplete. The following preprocessing steps were systematically applied:

- Dimensionality Reduction: The id and dataset/origin columns were dropped as they carry no predictive clinical value.
- Imputation of Missing Values: Missing categorical values were imputed using the statistical mode, while continuous numerical variables were imputed using the median to ensure robustness against outliers.



Missing Percentage
Figure 1: Percentage of missing values across the dataset features prior

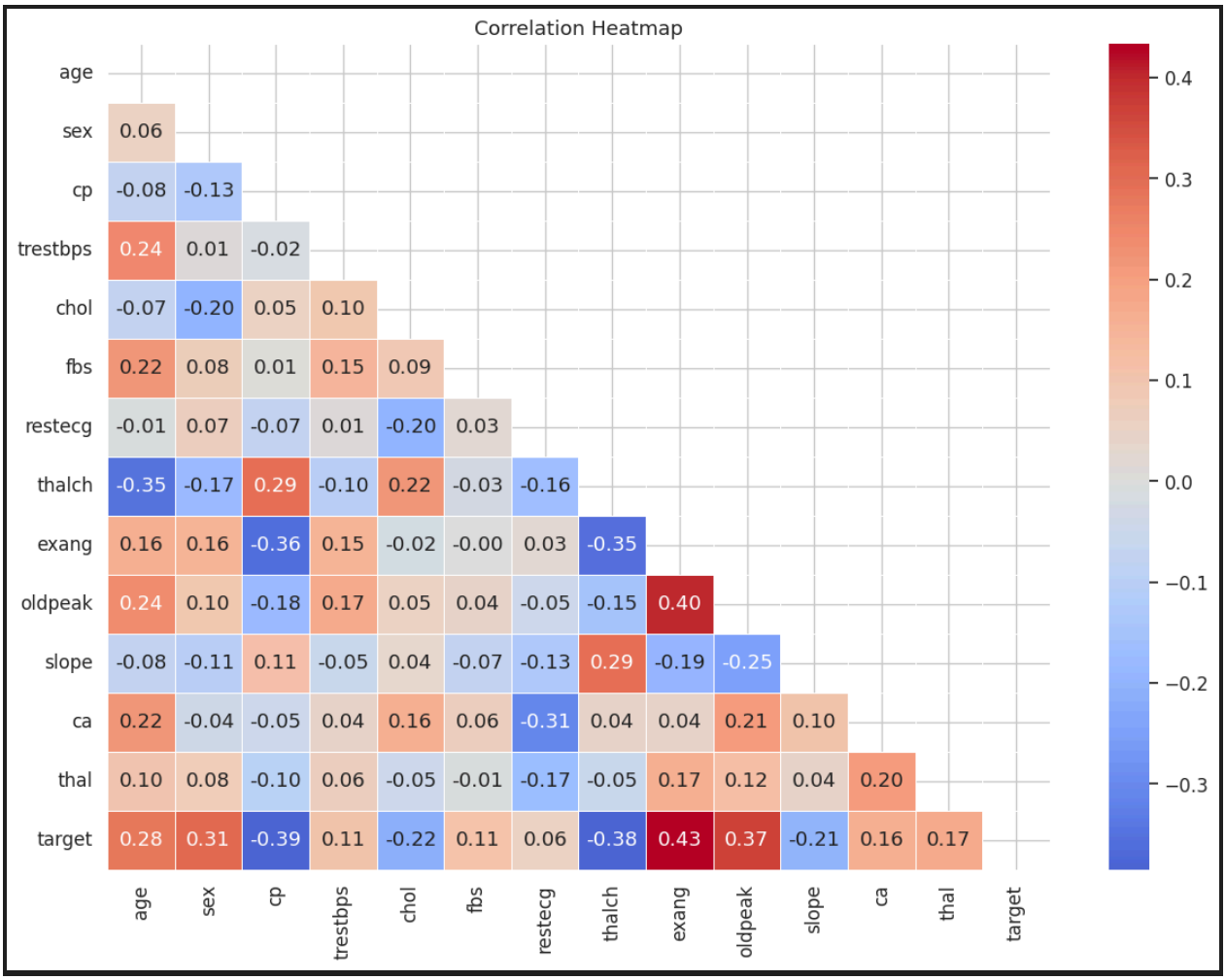
- Target Engineering: The raw num variable (ranging from 0 to 4) was transformed into a binary classification target where 0 indicates no disease and >0 indicates the presence of disease.



Target Distribution
Figure 2: Distribution of the engineered binary target variable showing a relatively balanced dataset.

4.3 Exploratory Data Analysis(EDA)

Comprehensive EDA was performed to uncover relationships between clinical features and the target variable.



Correlation Heatmap

Figure 3: Correlation heatmap revealing the linear relationships between continuous variables and the target.

Distributions by Target
Figure 4: Distributions of key features categorized by the target variable, illustrating the discriminatory power of features like maximum heart rate (thalach) and age.

4.4 Feature Engineering

To prepare the data for the neural network, rigorous encoding and scaling techniques were applied:

- Encoding: Binary variables (sex, fbs, exang) were mapped to 1/0. Nominal categorical variables (cp, restecg, slope, thal) were processed using Scikit-Learn's LabelEncoder.
- Outlier Treatment: The Interquartile Range (IQR) method was employed to cap outliers in continuous variables (trestbps, chol, thalach, oldpeak), preventing extreme values from skewing the model weights.

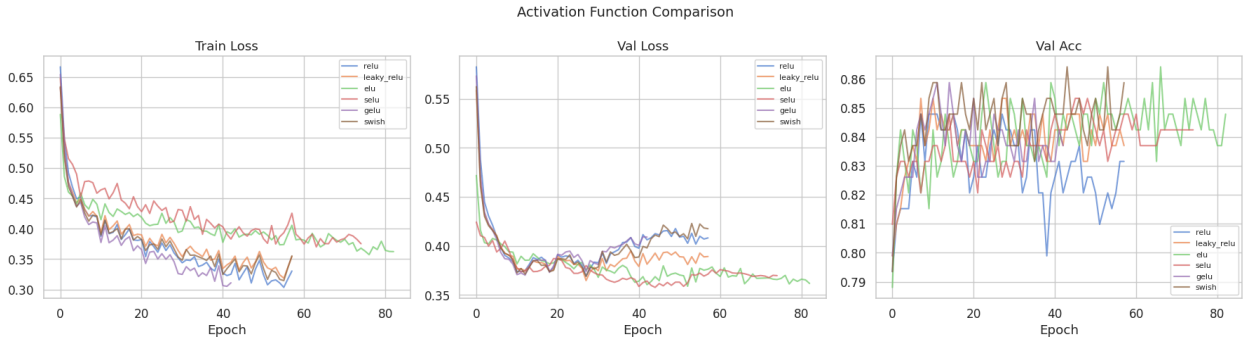
- Scaling: A StandardScaler was applied to all feature vectors, ensuring a mean of 0 and variance of 1, which is critical for the convergence of gradient descent in deep learning.

5. Model Architecture and Experimentation

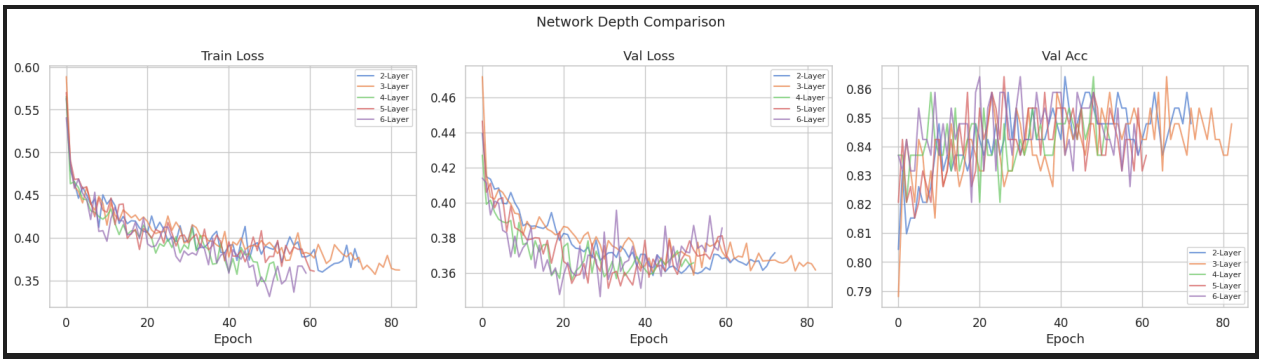
Instead of relying on classical ML algorithms (SVM, Random Forest), the study focused exclusively on Deep Neural Networks (DNN) using PyTorch. Extensive hyperparameter tuning was conducted through isolated experiments.

5.1 Activation Function and Depth Experiments

- Activation Functions: ELU outperformed ReLU, LeakyReLU, and Swish.
- Network Depth: Architectures ranging from 2 to 6 layers were tested. A 3-layer architecture [128, 64, 32] yielded the optimal balance between underfitting and overfitting.



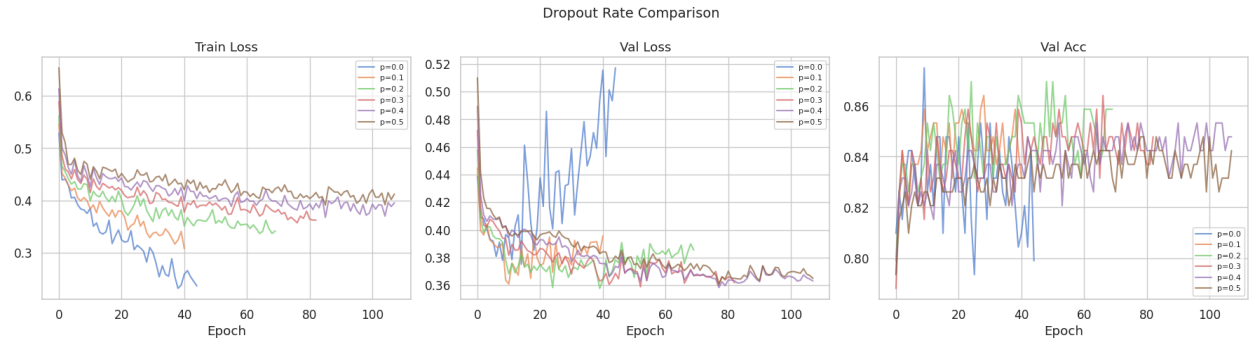
Activation Function Comparison
Figure 5: Training and validation curves comparing various activation functions.



Network Depth Comparison
Figure 6: Impact of network depth on convergence and validation accuracy.

5.2 Regularization and Optimization

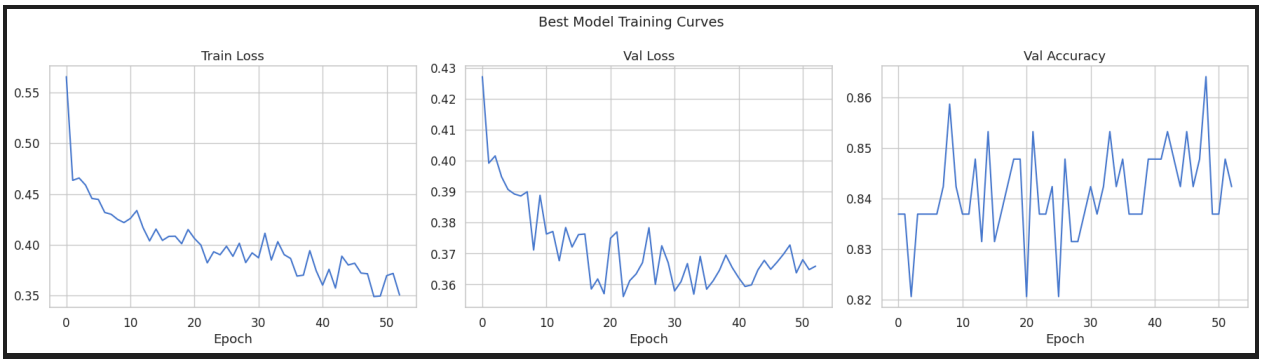
- Regularization: Various dropout rates were tested. A dropout rate of 0.3 combined with Batch Normalization (BatchNorm1d) proved most effective in generalizing the model and preventing overfitting.
- Optimization: The Adam optimizer with a learning rate of 0.01 and a ReduceLROnPlateau learning rate scheduler facilitated rapid and stable convergence.



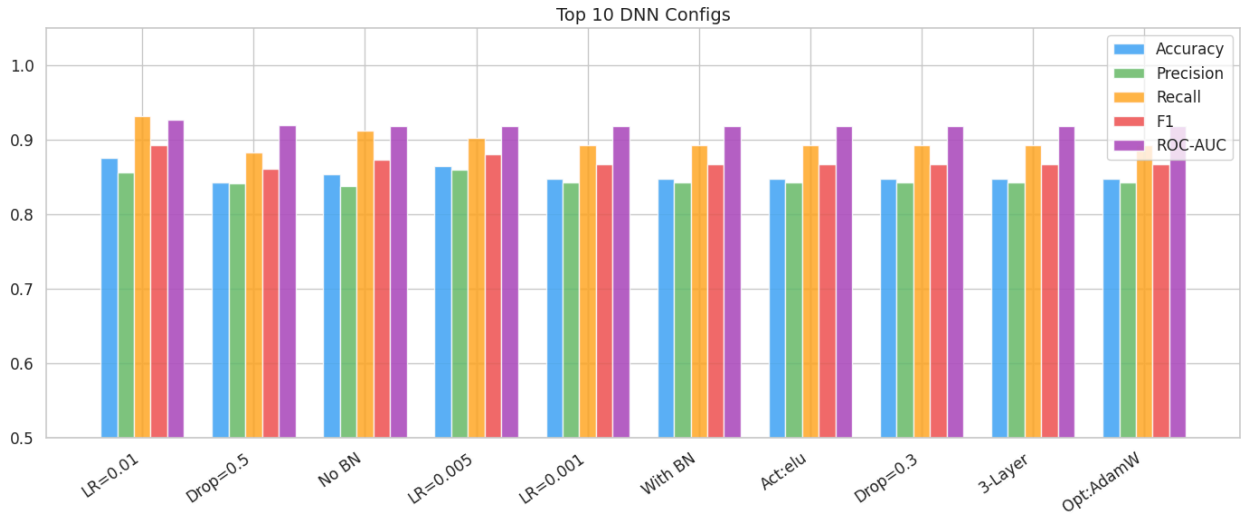
Dropout Rate Comparison
Figure 7: Effect of different dropout rates on the validation loss and accuracy.

5.3 Optimal Model Training

The final optimal configuration was trained over 400 epochs, demonstrating smooth convergence.



Best Model Training Curves
Figure 8: Training loss, validation loss, and validation accuracy for the final optimal DNN.



Top 10 Configs
Figure 9: Comparison of the top 10 neural network configurations based on multiple classification metrics.

6. Software Implementation

6.1 Backend Implementation

To operationalize the model, a REST API was developed using Python's Flask framework.

- The optimal model weights (.pth), fitted StandardScaler, and LabelEncoders were exported from the training script (train_and_export.py).
- The app.py script serves a /api/predict POST endpoint. It ingests raw JSON patient data, dynamically applies the exact preprocessing pipeline (encoding, clipping, scaling) used during training, and passes the tensor to the PyTorch model for inference.
- The API calculates and returns a risk probability mapping and a stratified risk level (Low, Moderate, High, Critical).

6.2 Frontend Implementation

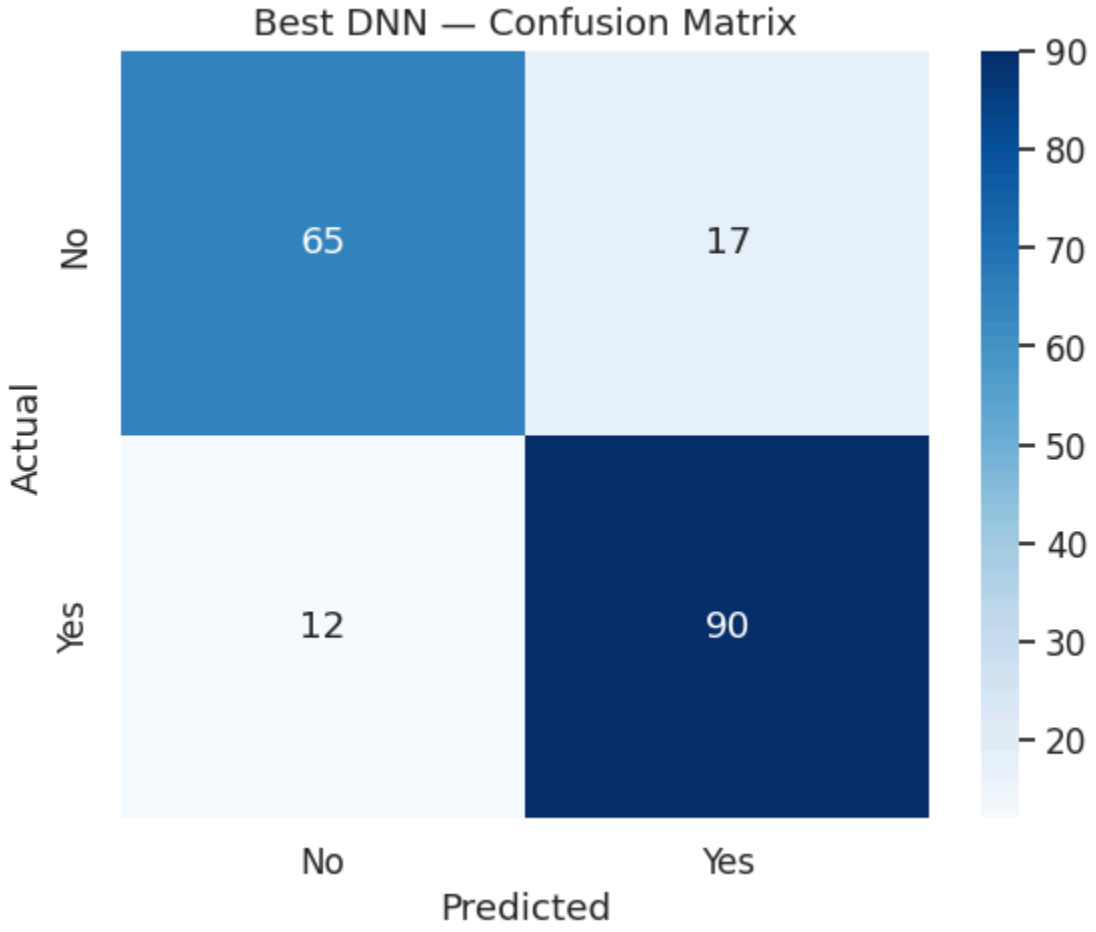
A premium, interactive web interface was constructed using React and Vite.

- Aesthetic: Designed using a modern dark-mode glassmorphism paradigm (translucent cards, blurred backgrounds, deep medical navy hues) to provide a clinical yet state-of-the-art feel.
- User Input: A structured, responsive form categorizes the 13 clinical inputs into logical groups (Demographics, Vitals, Symptoms, Labs, Diagnostics).
- Visualization: The Result Panel utilizes dynamic SVG circular progress rings and gradient risk bars to visually communicate the prediction probability and severity to the user in real-time. This provides immediate, interpretable feedback based on the DNN output.

7. Results and Evaluation

The final Deep Neural Network model was evaluated on a stratified 20% hold-out test set. The model demonstrated exceptional predictive capabilities:

- Accuracy: 87.50%
- Precision: 85.58%
- Recall (Sensitivity): 93.13%
- F1-Score: 89.20%
- ROC-AUC Score: 92.66%



Confusion Matrix
Figure 10: Confusion Matrix for the final Deep Neural Network, highlighting a low false-negative rate.

The remarkably high recall (93.13%) is particularly crucial in a medical context. Minimizing false negatives (failing to diagnose a sick patient) is the primary objective of

any clinical diagnostic support system, ensuring that patients requiring intervention are not overlooked.

8. Conclusion

The CardioAI project successfully demonstrates the viability of utilizing deep neural networks for cardiovascular disease prediction. By systematically addressing data inconsistencies, rigorously tuning a PyTorch DNN, and deploying the system through a full-stack architecture, we have created a robust, end-to-end diagnostic tool.

The resulting application not only achieves high statistical accuracy (92.66% AUC) but also provides a highly intuitive and visually engaging interface, bridging the gap between raw data science and practical healthcare technology. The implementation highlights how machine learning can serve as a powerful supplementary tool in clinical settings. Future enhancements could include the integration of larger, more diverse datasets and the implementation of model explainability techniques (such as SHAP values) to provide clinicians with transparent, feature-level reasoning behind each prediction.